In [1]:
```python
import warnings
warnings.filterwarnings("ignore")
import os, pandas as pd, matplotlib.pyplot as plt, numpy as np, seaborn as sns, glob
```

In [2]:
```python
df = pd.read_csv('FRs_and_gaze_P54CS_R1.csv')
df.head()
```

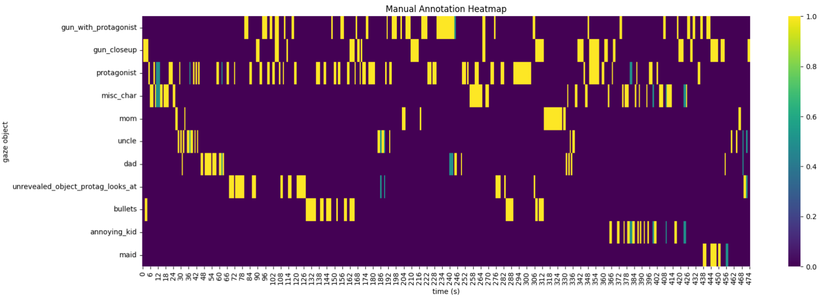
Out[2]:

	frame	gun_with_protagonist	gun_closeup	protagonist	misc_char	mom	uncle	dad	unrevealed_object_protag_looks_at	bullets	...	vmPFC_neur83	vmPFC_neur84	vmPFC_neur85	vmPFC_neur86	vmPFC_neur87	vmPFC_neur
0	11	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.898057	3.761152	-1.321287	0.537737	1.087061	0.5335
1	12	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	-0.296837	0.870299	1.743970	-0.792602	-0.941835	2.8208
2	13	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	...	-0.296837	0.870299	-0.554973	0.537737	-0.941835	1.4484
3	14	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	...	-0.296837	-1.056936	0.211341	-0.792602	0.072613	0.9909
4	15	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.300610	0.870299	-1.321287	-0.792602	-0.941835	1.4484

5 rows × 104 columns

plotting objects gazed at

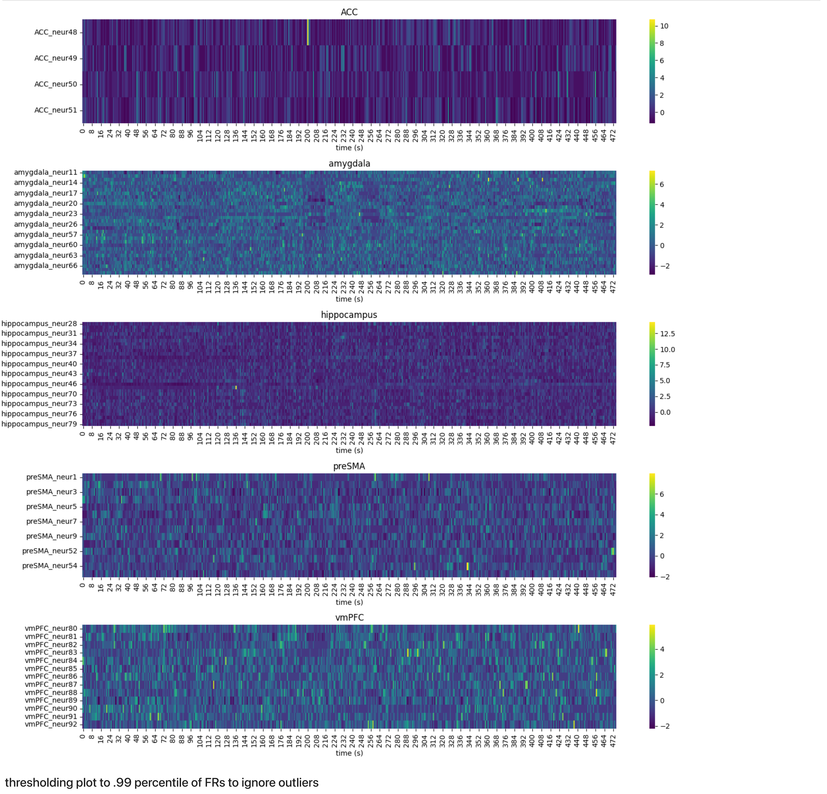
In [3]:
```python
gaze_cols = [col for col in df.columns if 'neur' not in col and 'frame' not in col]
df_gaze = df[gaze_cols]

plt.figure(figsize=(18, 6))
sns.heatmap(df_gaze.T, cmap='viridis', cbar=True)
plt.xlabel('time (s)')
plt.ylabel('gaze object')
plt.title('Manual Annotation Heatmap')
plt.tight_layout()
plt.show()
```



plotting boxcar normed FRs

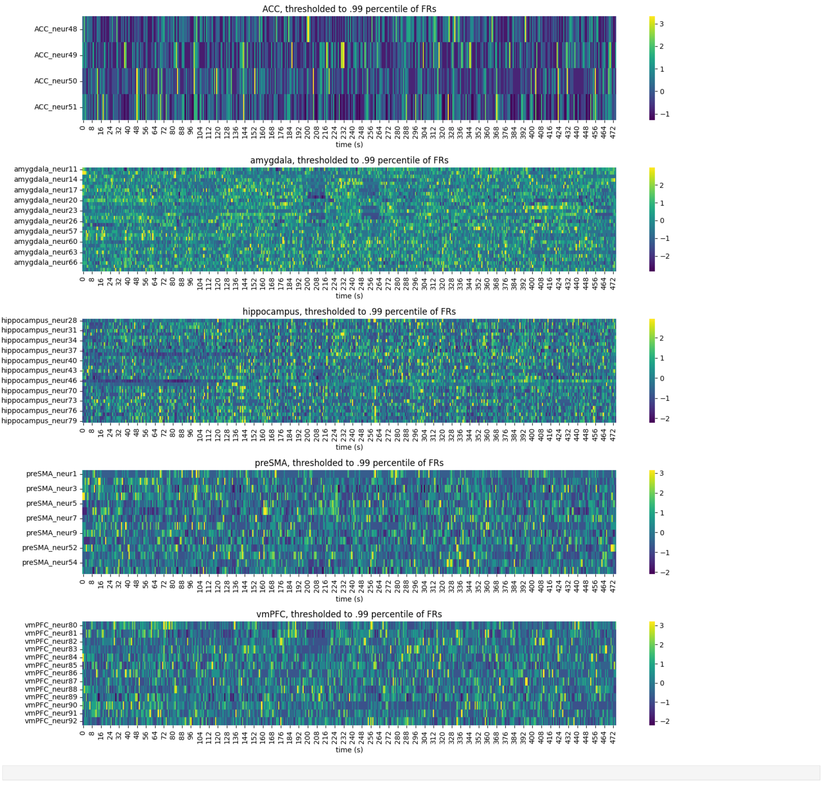
note that a few outliers make it difficult to see neuronal variation. see more in the next block.

In [4]:
```python
# get all region names from neuron col names
neuron_cols = [col for col in df.columns if '_neur' in col]
regions = sorted(set([col.split('_neur')[0] for col in neuron_cols]))

fig, axs = plt.subplots(5, 1, figsize=(15, 15))
axs, ax = axs.flatten(), 0

for region in regions:

    # grab columns for the current region
    region_cols = [col for col in df.columns if col.startswith(region)]
    df_time_by_neur = df[region_cols]

    sns.heatmap(df_time_by_neur.T, cmap='viridis', cbar=True, ax=axs[ax])
    axs[ax].set(xlabel='time (s)', title = region)
    ax += 1

plt.tight_layout()
plt.show()
```



thresholding plot to .99 percentile of FRs to ignore outliers

In [5]:
```python
fig, axs = plt.subplots(5, 1, figsize=(15, 15))
axs, ax = axs.flatten(), 0

for region in regions:

    # grab columns for the current region
    region_cols = [col for col in df.columns if col.startswith(region)]
    df_time_by_neur = df[region_cols]

    sns.heatmap(df_time_by_neur.T, cmap='viridis', cbar=True, ax=axs[ax], vmax = np.percentile(df_time_by_neur.values, 99))
    axs[ax].set(xlabel='time (s)', title = f'{region}, thresholded to .99 percentile of FRs')
    ax += 1

plt.tight_layout()
plt.show()
```



In [ ]: